higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

WELDING
NQF LEVEL 4

12 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1

1.1.1	D
1.1.2	A
1.1.3	G
1.1.4	J
1.1.5	B
1.1.6	L
1.1.7	C
1.1.8	E
1.1.9	F
1.1.10	H

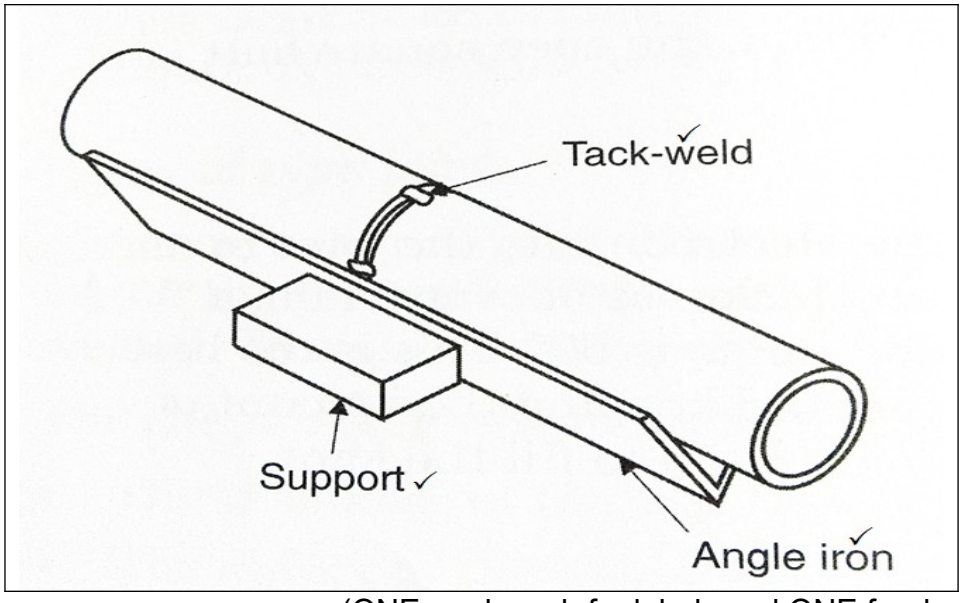
(10 × 1) (10)

1.2

A	Power Supply\Welding machine
B	Earth cable
C	Earth clamp
D	Work bench
E	Electrode\welding rod
F	Electrode holder

(Any relevant 6 × 1) (6)

1.3



(ONE mark each for labels and ONE for drawing.) (4)

[20]

QUESTION 2

2.1
- Flat rotated/horizontal rotated.
- Flat fixed/Horizontal fixed.
- Multiple not rotated/Vertical position.
- Multiple not rotated/Inclined position. (4)

2.2
- The electrode must deliver a stable arc.
- Must be able to produce an even run.
- Minimum spatter should result.
- Produce maximum welding strength.
- Slag must be easily removed. (5)

2.3 Welding starts when an electric arc✓ is struck between the tip of the electrode and the work piece.✓ The intense heat of the arc melts the tip of the electrode and the surface of the work piece close to the arc.✓ Tiny globules of molten metal rapidly form on the tip of the electrode,✓ and then transfers trough the arc stream into the molten weld pool.✓ The filler metal is deposited as the electrode is consumed.✓ (6)

2.4
- Spatter
- Deformation
- Arc striking difficulties
- Porosity
- Lack of root penetration
- Undercutting
- Arc length to long (Any relevant 5 × 1) (5)

[20]

QUESTION 3

3.1
- AC main supply
- Welding machine
- Welding cable
- Electrode holder
- Electrode\welding rod
- Shielding gas
- Trip switch
- Return cable
- Current control (Any relevant 6 × 1) (6)

3.2
3.2.1 inert gas welding
3.2.2 non-consumable
3.2.3 inert gas
3.2.4 contamination
3.2.5 odourless

(5 × 1) (5)

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3.3 'Direct current electrode positive (DCEP)', works opposite to DCEN, the torch is connected to the positive terminal✓ while the earth clamp is connected to the negative terminal.✓ The electrode is on the positive side of the arc.✓ The electrode is larger.✓ (4)

3.4
- Do not drop cylinders.
- Chain cylinders.
- Do not allow cylinders to bang each other.
- Do not overload the vehicle.
- Use safety equipment when loading or off-loading cylinders.
- Do not roll the cylinders
- Ensure the vehicle is properly cage up and the cylinders are secured before transporting.
- Do not drag the cylinders on the floor when transporting.

(Any relevant 5 × 1) (5)

[20]

QUESTION 4

4.1
A	Collet holder
B	Nozzle
C	Torch body
D	Electrode
E	O-ring
F	Cap

(6)

4.2
- High quality weld
- All welding positions
- No slag
- No spatter
- Joins dissimilar metals (5)

4.3 The electrode is not consumed✓ during Gas Tungsten Arc Welding, because tungsten has the highest melting point✓ of all pure metals.✓ The electrode does not have to be replaced.✓ (4)

4.4
- Choose the correct torch for the current range.
- Ensure that the electrode chosen is correct.
- Use the correct shape of electrode.
- Set the electrode so that it extends between 3-5 mm.
- Check all electrical connections.
- Ensure the equipment is in good working condition.

(Any relevant 5 × 1) (5)

[20]

QUESTION 5

5.1
- Use the correct PPE.
- Work within the design parameters of the equipment.
- Ensure all equipment are checked and serviced according to the manufactures specification.
- Use the correct gas.
- Ensure the correct current and gas pressures are used.
- Ensure there is sufficient ventilation.
- Ensure there is adequate lighting.
- Ensure that shields are used to protect fellow workers against the arc.

(Any relevant 6 × 1) (6)

5.2 Radiographic and ultrasonic welding inspections are non-destructive testing.✓ They are used to examine the internal structures of welds.✓ Radiographic method used x-ray film.✓ Areas of the weld were the thickness changes due to porosity or cracks will appear on the film.✓ (4)

5.3
- Equipment is expensive.
- Equipment requires more maintenance.
- The welding gun is difficult to get into small spaces.
- Not suitable for welding in windy conditions.
- Require a high operation skill. (5)

5.4
- Poor gas shielding.
- Wire speed to high.
- Dirty work piece.
- Dirty welding wire.
- Poor welding condition (windy area).
- Current set to high on the welding machine.
- Arc length to long.
- Arc force set to high. (Any relevant 5 × 1) (5)

[20]

TOTAL: 100

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